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Freight Rate Prediction

Machine Learning Engineer Assessment

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A reproducible solution for estimating a load's posted rate from route, equipment, weight, date, and supplied market signals. The final deliverables contain predictions for every unlabeled load and every fixed December scenario.

Reserved main-model MAE	Baseline MAE	MAE reduction
$116.19	$229.10	49.3%

Internal out-of-time evaluation: 9,523 September-October loads; models fitted through August. Source: [S3].

Data and prediction task

Dataset	Rows	Period / role
Labeled development	48,000	January-October 2025; target: posted_rate
Final inputs	12,000	November-December 2025; no labels supplied
Fixed chart inputs	31	One scenario for every day of December 2025

Data-quality findings and handling

Issue	Labeled	Final	Treatment
Missing weight	300	165	Keep unknown; preserve missingness flag
Negative weight	292	145	Set unknown; preserve invalid-value flag
Missing market index	374	249	Native numeric missing values plus indicator
Unseen city	-	1,447 loads	Unknown-safe model; separate city stress test

IDs are unique and no duplicate feature rows were found. Supplied coordinates are within valid ranges and consistent per city, but were not externally geocoded. Extreme rate labels are retained; a large rate alone is not evidence of an error.

Source: [S1]. Eight pickup and delivery city names are absent from labeled training. Missing and negative weight counts are disjoint.

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Validation and features

Match the future-prediction horizon and keep fitted transformations within training folds.

Purpose	Train	Validate	Train / validate rows
Development 1	Jan-Apr	May-Jun	19,110 / 9,696
Development 2	Jan-Jun	Jul-Aug	28,806 / 9,671
Reserved check	Jan-Aug	Sep-Oct	38,477 / 9,523
Final refit	Jan-Oct	Nov-Dec (unlabeled)	48,000 / 12,000

The first two expanding windows provide 19,367 nonoverlapping, out-of-time validation loads for model selection. MAE in dollars is the development selection metric; RMSE and R-squared are also reported because the employer's hidden metric is unspecified. The two-month horizons mirror the final prediction period.

Model roles and fixed hyperparameters were recorded before inspecting September-October model performance. No early stopping, calibration, feature selection, or model switching used that window. Dataset-level exploration included all labeled months: this is a reserved performance check, not an externally sequestered blind test.

Leakage and preprocessing controls

The stateless cleaner does not learn from labels. It excludes load_id and posted_rate, marks nonpositive weights unknown, and adds separate missing/invalid flags. Ridge imputation, scaling and one-hot encoding fit on each training fold only. CatBoost and equipment medians likewise fit only on training rows. No evaluation rows or difficult labels are dropped.

Two explicit inference paths

Path	Features
Main model	Pickup/delivery/equipment categories; distance; cleaned weight and flags; supplied coordinates, coordinate deltas and derived coordinate distance; calendar fields; market index and quote signal
December chart	Only pickup, delivery, distance, equipment, weight and date-derived/weight-quality features. Coordinates, market index and quote signal are excluded during both fitting and inference.

Calendar inputs include month, weekday, day of month, day of year and weekend. The chart path requires no fabricated future signals or external city lookup. Native categorical handling supports unfamiliar cities; a separate development stress test measures that risk.

Sources: [S1], [S2], [S4], [S5]. Implementation: freight/data.py, features.py, splits.py, models.py and model_policy.json.

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Model selection

Compare simple baselines, objectives and signal availability before the reserved check.

Candidate	MAE ($)	RMSE ($)	R-squared
CatBoost full - MAE	140.70	635.20	0.8253
CatBoost market - MAE	146.67	634.00	0.8259
CatBoost core - MAE	160.37	638.11	0.8237
CatBoost chart - MAE	161.55	638.35	0.8236
CatBoost core - RMSE	198.02	671.42	0.8048
Ridge core rate / mile	213.18	651.35	0.8163
Equipment rate / mile	214.45	658.54	0.8122
Median rate	1,152.07	1,574.79	-0.0738

Pooled development windows, 19,367 loads per candidate. Source: [S2]. Core excludes both market index and quote signal; market adds only market index; full adds both.

The full-feature MAE model has the lowest development MAE ($140.70), improving 34.4% over the equipment baseline. MAE training is less sensitive to extreme labels than squared-error training. The market-only model has slightly lower RMSE, so the tradeoff is retained rather than claiming one candidate dominates every metric.

Signal stability and geographic robustness

The quote signal's correlation with actual rate per mile changes sign across months. Its overall correlation hides that change. This motivates explicit signal ablations; it is not proof of leakage. Quote-time availability and signal generation are undocumented.

City stress candidate	Unseen-city MAE ($)
CatBoost market - MAE	170.81
CatBoost full - MAE	174.35
CatBoost core - MAE	192.23
CatBoost chart - MAE	198.76
Equipment rate / mile	254.33

Eight cities were chosen by stable, target-independent hashes from the first training window. Removing both route-role occurrences from training produces 4,403 unseen-city evaluation loads across the two windows. The market-only edge is small and comes from one city subset; it does not justify extra city routing. The full model remains the predeclared main model.

Sources: [S1], [S2], [S4]. All CatBoost candidates: 500 iterations, depth 6, learning rate 0.05, L2 leaf regularization 5, seed 42, four threads; no validation early stopping.

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Reserved evaluation

Frozen configurations evaluated on all September-October loads, without another tuning round.

Role	MAE ($)	RMSE ($)	R-squared
Equipment baseline	229.10	670.66	0.8069
Main model	116.19	636.24	0.8262
Reduced-input chart model	113.09	635.14	0.8268

N = 9,523. Training N = 38,477. Source: [S3]. The main-model MAE reduction is 49.3% versus the predeclared reference.

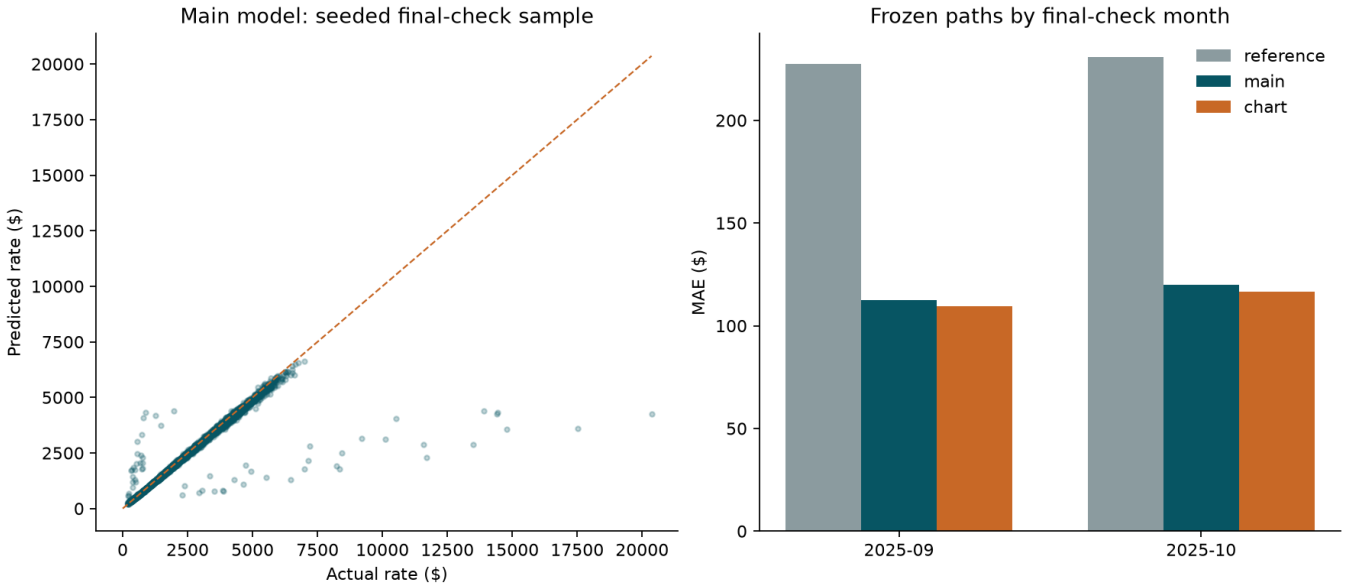


Figure 1. Reserved-check main-model predictions (seeded sample) and monthly MAE. Source: [S3]. The diagonal marks perfect predictions.

Interpretation and error concentration

The chart model's slightly lower reserved MAE does not trigger a role switch: the full model was selected on development data and the chart model was selected for reduced-input availability. Both fixed configurations are retained for full-data refitting.

In the earlier development predictions, median absolute error is $58.29, while the largest 1% of errors contribute 95.2% of total squared error. This explains why RMSE is much larger than MAE. Every large error remains in the headline scores; tail removal is not used to improve reporting.

The reserved temporal window contains no unseen cities. The separate city stress results therefore complement, rather than replace, the chronological check. These internal results do not measure accuracy on Spotter's unlabeled final set.

Sources: [S3], [S4]. Metric arithmetic and source labels were independently checked against the saved row-level predictions.

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Fixed December predictions

A separate, validated reduced-input path for the exact scenario supplied with the assessment.

Fixed field	Value
Pickup / delivery	Lexington / Fort Wayne
Distance / equipment / weight	360 miles / Dry Van / 32,000 lb
Dates	December 1-31, 2025; only date changes

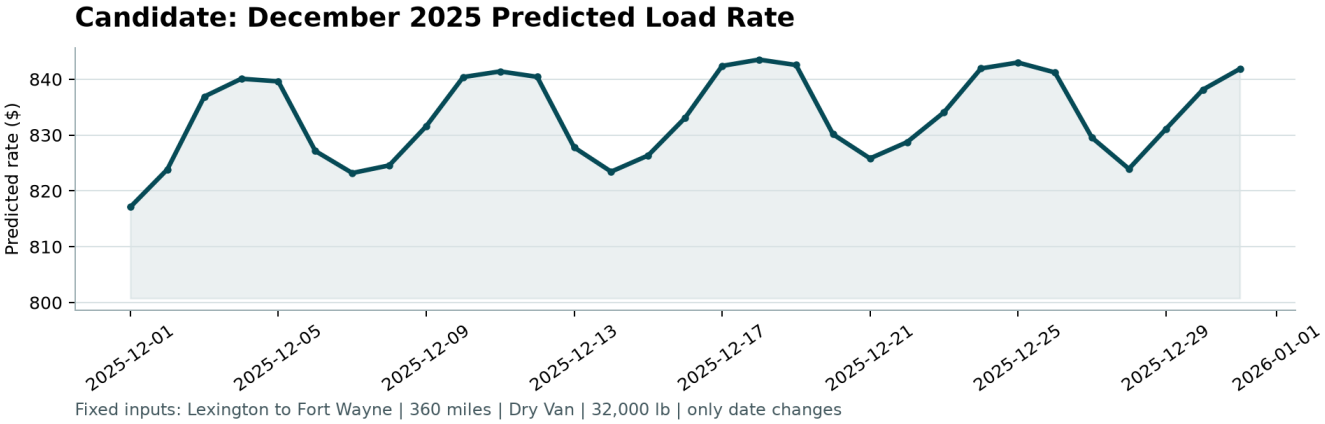


Figure 2. Exact candidate_december.png generated by the unchanged supplied score.py. The image is embedded without redrawing or modifying the predictions. Source: [S6].

Across the 31 fixed scenarios, predicted rates range from **$817.10** to **$843.51**, with mean **$833.38**. These are model outputs, not observed future prices.

Input availability and inference contract

The scenario file does not supply pickup/delivery coordinates, market index or quote signal. The chart model excludes those inputs throughout training and temporal evaluation. It uses the same labeled training rows and frozen hyperparameters, with the reduced feature set. The final chart model is fitted on all 48,000 labeled loads.

The completed file preserves all six original input fields, the seven-column schema, and the original row order. Only predicted_rate is filled. The supplied scorer validates one finite positive prediction for each December date and generates the required chart.

Scope of the curve

Date-derived splits can produce weekday and within-month variation, but one partial year does not establish annual seasonality, holiday effects or a calibrated December forecast. No holiday uplift, smoothing or manual date-specific price adjustment is added. The chart path need not equal full-model predictions for similar loads with more inputs.

Sources: december-chart-inputs.csv, december_predictions.csv, [S3], [S5], [S6].

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Reproducibility and delivery

Code, pinned dependencies, native models and independently checked prediction contracts.

From an activated Python 3.12 environment at the repository root:

```
python -m pip install -r requirements-lock.txt
python -m unittest discover -s tests -v
python -m freight.submission
python -m freight.audit --require-saved-models
```

The generation command fits the unchanged main/chart configurations on all labeled rows, saves native CatBoost models, reloads them, verifies every prediction, fills both CSVs by their contracts, and runs the employer's scorer. Prediction-only mode reuses matching saved models. Input, code, policy, output and model hashes are recorded in manifests.

A clean source copy and independently installed Windows environment passed all 26 tests, refitted both models from scratch, and reproduced both CSVs and the chart byte for byte. Eleven local integrity checks pass. Hosted Windows/Linux dependency, test, artifact-audit and scorer checks passed; full Linux refit byte identity is not claimed.

Limitations

Final labels and the grading metric are unavailable. Signal relationships shift, the geographic test uses one fixed city subset, and heavy rate tails dominate RMSE. No prediction intervals or causal claims are made. Reproductions in another environment retain the frozen configuration and preserve the original evaluation evidence, but exact values across every platform are not guaranteed.

Artifacts and evidence

Reference	Repository source
S1 - Data audit	reports/eda/audit.json; reports/eda/exploration.md
S2 - Model comparison	reports/benchmark/leaderboard.csv; fold_metrics.csv
S3 - Reserved evaluation	reports/final_check/metrics.csv; segment_metrics.csv; manifest.json
S4 - Robustness	reports/robustness/location_pooled_metrics.csv; residual_summary.csv
S5 - Frozen decisions	MODEL_DECISIONS.md; model_policy.json
S6 - Final artifacts	validation_predictions.csv; december_predictions.csv; scorer_results/candidate_december.png
S7 - Integrity evidence	reports/audit/checks.json; clean_reproduction.json; reports/submission/manifest.json

Sources are supplied assessment data or generated results in the repository; no external data augments training. All currency values are dollars.

Repository: github.com/AffanRM/spotter_ml

Submitted prediction files: validation_predictions.csv and december_predictions.csv. Camera-on Loom walkthrough is supplied separately by the candidate.